

ISFAHAN UNIVERSITY OF TECHNOLOGY

Department of Mechanical Engineering

STABILITY ANALYSIS OF A FOUR-BAR PARALLEL SYSTEM CONNECTED BY SPRINGS

Advanced Dynamics Course Project

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Abstract

This project investigates the equations of motion and stability of a four-degree-of-freedom mechanical system consisting of four slender rods connected in parallel by elastic springs and viscous dampers. The nonlinear equations of motion are formulated using four approaches: Newton's method, the Lagrange formulation, the direct Hamilton principle, and the canonical Hamilton formulation. The resulting dynamic response is evaluated for a prescribed initial condition. The system is then linearized about equilibrium configurations and its small-deviation stability is examined from the eigenvalues of the state matrix. Finally, large-deviation behavior is studied using Lyapunov's direct method, with the total mechanical energy used as a candidate Lyapunov function and an estimate of the region of attraction obtained from its level sets.

The numerical parameters used in the original project are $m = 0.5$ kg, $M = 0$, $k = 0.75$, $L = 0.5$ m, $g = 9.81$ m/s², and $c = 0.1$. For the reported initial condition, $\theta_1 = \theta_2 = \theta_3 = 0$, $\theta_4 = \pi/6$, with zero initial angular velocities, the computed responses decay toward the equilibrium configuration.

Project Objectives

- Derive the nonlinear equations of motion for the four-degree-of-freedom system.
- Verify the formulation using Newton, Lagrange, direct Hamilton, and canonical Hamilton approaches.
- Compute the dynamic displacement and angular-velocity responses for a specified initial condition.
- Assess small-deviation stability about the $\theta = 0$ and $\theta = \pi$ equilibrium configurations.
- Construct a Lyapunov function and examine large-deviation stability and the corresponding region of attraction.

Nomenclature

Symbol	Definition
m	Mass of each rod
I	Moment of inertia of each rod
L	Rod length
T	Total kinetic energy
k	Spring stiffness coefficient
U	Total potential energy
c	Viscous damping coefficient
C	Damping coefficient notation used in the source formulation
M	Applied torque on rod 4
L	Lagrangian
H	Hamiltonian
p	Generalized momentum
V	Lyapunov function
δ	Variational operator
W_{nc}	Work of non-conservative forces
U_g	Gravitational potential energy
U_e	Elastic potential energy
g	Gravitational acceleration
$\theta_1-\theta_4$	Angular displacements of rods 1–4
$\dot{\theta}_1-\dot{\theta}_4$	Angular velocities of rods 1–4

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1. Equations of Motion

1.1 Newton's Method

The physical system has four degrees of freedom. Each rod is modeled as a slender rigid link of length L and mass m , with the rods connected consecutively by linear springs. Viscous damping is included, and an external torque M is applied to the fourth rod. The kinetic diagram is used to write the rotational balance about each fixed pivot.

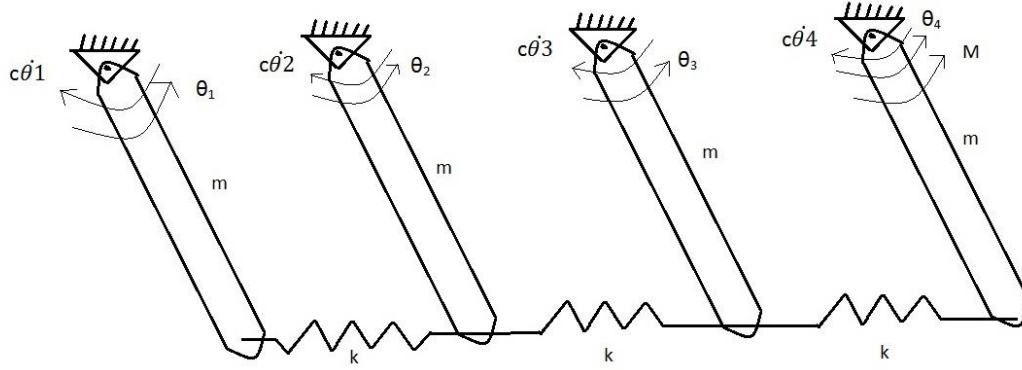


Figure 1. Physical model of the four-degree-of-freedom system.

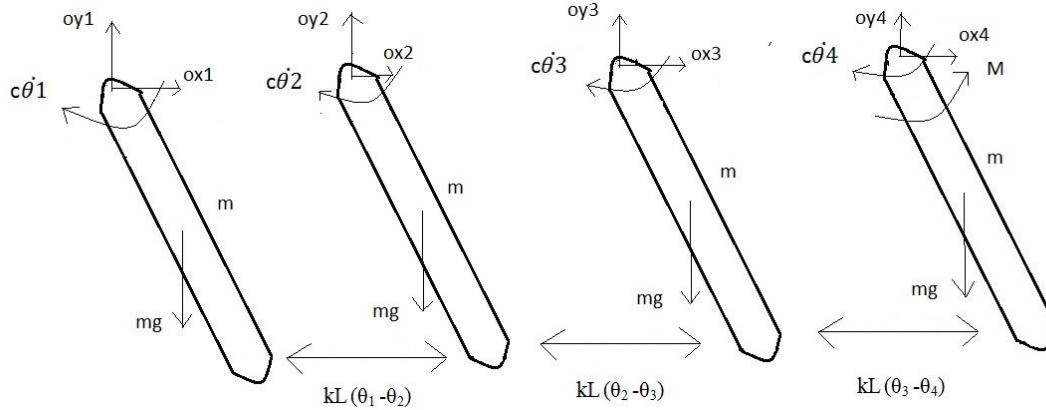


Figure 2. Kinetic diagram of the four-degree-of-freedom system.

For a slender rod rotating about one end, the mass moment of inertia is

$$I_0 = I_G + m(L/2)^2 = (mL^2/12) + (mL^2/4) = mL^2/3.$$

For small angular displacements, the spring extensions are approximated by

$$x_1 = L \sin(\theta_1) \approx L\theta_1, \quad x_2 = L \sin(\theta_2) \approx L\theta_2, \quad x_3 = L \sin(\theta_3) \approx L\theta_3, \quad x_4 = L \sin(\theta_4) \approx L\theta_4.$$

The four nonlinear second-order equations of motion obtained from the rotational moment balances are

$$(1) \quad (mL^2/3) \ddot{\theta}_1 + c \dot{\theta}_1 + (m g L/2) \sin \theta_1 + kL^2(\theta_1 - \theta_2) = 0.$$

$$(2) \quad (mL^2/3) \ddot{\theta}_2 + c \dot{\theta}_2 + (m g L/2) \sin \theta_2 + kL^2(2\theta_2 - \theta_1 - \theta_3) = 0.$$

$$(3) \quad (mL^2/3) \ddot{\theta}_3 + c \dot{\theta}_3 + (m g L/2) \sin \theta_3 + kL^2(2\theta_3 - \theta_2 - \theta_4) = 0.$$

$$(4) \quad (mL^2/3) \ddot{\theta}_4 + c \dot{\theta}_4 + (m g L/2) \sin \theta_4 + kL^2(\theta_4 - \theta_3) = M.$$

These equations constitute the nonlinear dynamic model used throughout the remainder of the project.

1.2 Lagrange's Method

In the Lagrange formulation, the total kinetic and potential energies are first determined. Generalized non-conservative forces and viscous dissipation are then introduced into the governing equations.

$$L = T - U.$$

$$T = (mL^2/6)(\theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2).$$

$$U = U_g + U_e.$$

$$U = (mgL/2) \sum_{i=1}^4 (1 - \cos \theta_i) + (kL^2/2)[(\theta_1 - \theta_2)^2 + (\theta_2 - \theta_3)^2 + (\theta_3 - \theta_4)^2].$$

Rayleigh's dissipation function is

$$F = (c/2)(\theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2).$$

The generalized non-conservative forces are zero for rods 1–3, while the applied torque on rod 4 gives

$$Q_1 = Q_2 = Q_3 = 0, \quad Q_4 = M.$$

Using the Lagrange equations with Rayleigh dissipation

$$d/dt(\partial L/\partial \dot{\theta}_i) - \partial L/\partial \theta_i + \partial F/\partial \dot{\theta}_i = Q_i, \quad i = 1, \dots, 4,$$

the same four nonlinear equations listed in Eqs. (1)–(4) are recovered. This agreement provides a formulation check.

1.3 Direct Hamilton Principle

Hamilton's principle is applied in its direct variational form. The first variation of the action, together with the virtual work of non-conservative forces, is set to zero.

$$\delta \int (T - U) dt + \delta W_{nc} = 0.$$

The virtual variation of the kinetic energy, after integration by parts, produces the inertial terms. The potential-energy variation gives the gravitational and spring terms, while the non-conservative virtual work contains viscous damping and the applied torque M .

$$\delta W_{nc} = -c\theta_1\delta\theta_1 - c\theta_2\delta\theta_2 - c\theta_3\delta\theta_3 + (M - c\theta_4)\delta\theta_4.$$

Since $\delta\theta_1$, $\delta\theta_2$, $\delta\theta_3$, and $\delta\theta_4$ are arbitrary, the coefficients of the virtual displacements must vanish. The resulting equations are again Eqs. (1)–(4).

1.4 Canonical Hamiltonian Method

The canonical formulation introduces generalized momenta and the Hamiltonian function. The generalized momenta are

$$p_i = \partial L/\partial \dot{\theta}_i = (mL^2/3)\dot{\theta}_i, \quad i = 1, \dots, 4.$$

$$\dot{\theta}_i = 3p_i/(mL^2).$$

The Hamiltonian is defined as

$$H(q, p) = T + U.$$

$$H = \sum_{i=1}^4 [3p_i^2/(2mL^2)] + (kL^2/2)[(\theta_1 - \theta_2)^2 + (\theta_2 - \theta_3)^2 + (\theta_3 - \theta_4)^2] + (mgL/2)\sum_{i=1}^4 (1 - \cos \theta_i).$$

The dissipative generalized forces in canonical variables are

$$Q_{n,i} = -3cp_i/(mL^2), \quad i = 1,2,3; \quad Q_{n,4} = M - 3cp_4/(mL^2).$$

Hence, the first-order canonical state equations are

$$\theta_1 = 3p_1/(mL^2), \quad \theta_2 = 3p_2/(mL^2), \quad \theta_3 = 3p_3/(mL^2), \quad \theta_4 = 3p_4/(mL^2).$$

$$\dot{p}_1 = -kL^2(\theta_1 - \theta_2) - 3cp_1/(mL^2) - (m g L/2)\sin \theta_1.$$

$$\dot{p}_2 = -kL^2(2\theta_2 - \theta_1 - \theta_3) - 3cp_2/(mL^2) - (m g L/2)\sin \theta_2.$$

$$\dot{p}_3 = -kL^2(2\theta_3 - \theta_2 - \theta_4) - 3cp_3/(mL^2) - (m g L/2)\sin \theta_3.$$

$$\dot{p}_4 = M - kL^2(\theta_4 - \theta_3) - 3cp_4/(mL^2) - (m g L/2)\sin \theta_4.$$

2. Dynamic Response

The nonlinear second-order equations are converted to a first-order state-space representation and solved for the specified initial conditions. The original project uses MATLAB to obtain the displacement and angular-velocity histories.

2.1 System Parameters

Parameter	Value	Unit
m	0.5	kg
M	0	N·m
k	0.75	as used in the source model
L	0.5	m
g	9.81	m/s ²
c	0.1	as used in the source model

2.2 Initial Condition

$$\theta_1(0) = \theta_2(0) = \theta_3(0) = 0, \quad \theta_4(0) = \pi/6 = 30^\circ.$$

$$\dot{\theta}_1(0) = \dot{\theta}_2(0) = \dot{\theta}_3(0) = \dot{\theta}_4(0) = 0.$$

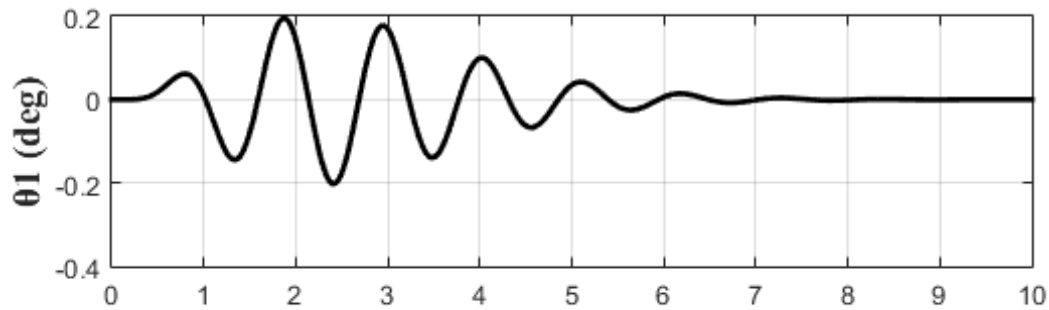


Figure 3. Displacement response of rod 1 under the specified initial condition.

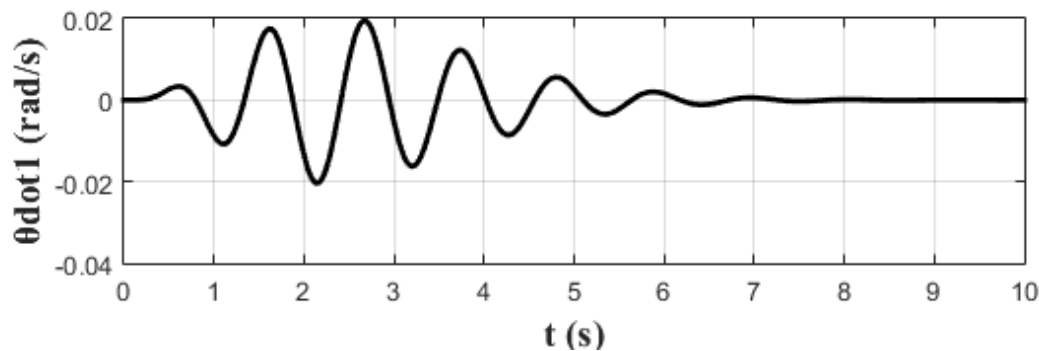


Figure 3b. Angular-velocity response of rod 1.

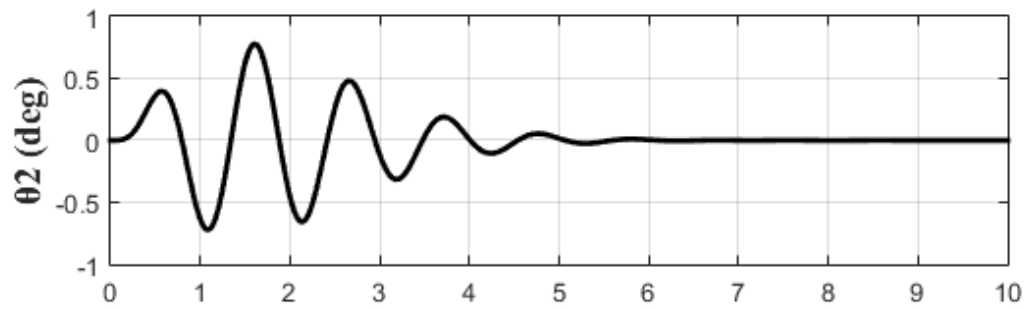


Figure 4. Displacement response of rod 2.

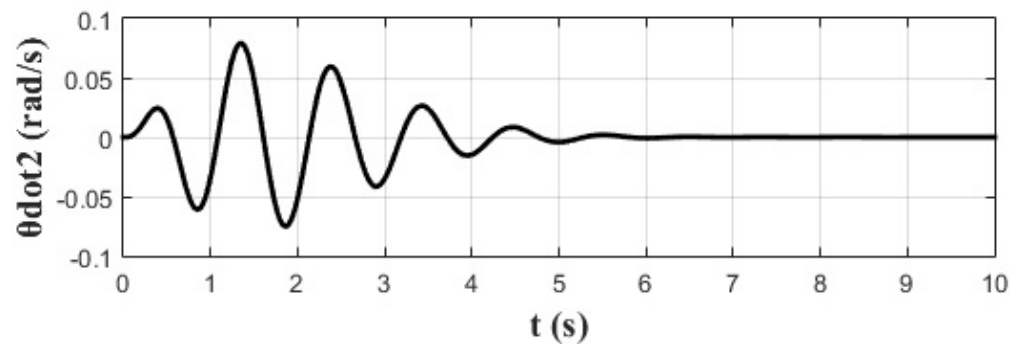


Figure 4b. Angular-velocity response of rod 2.

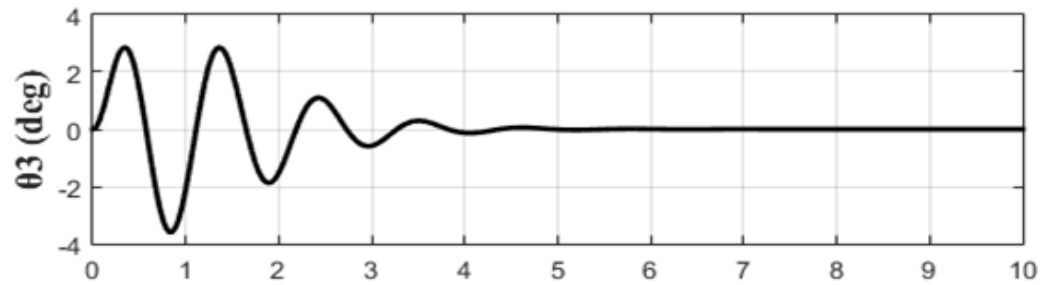


Figure 5. Displacement response of rod 3.

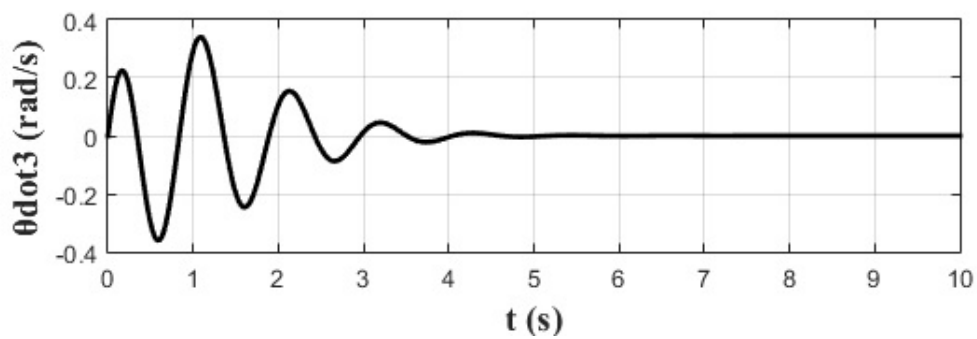


Figure 5b. Angular-velocity response of rod 3.

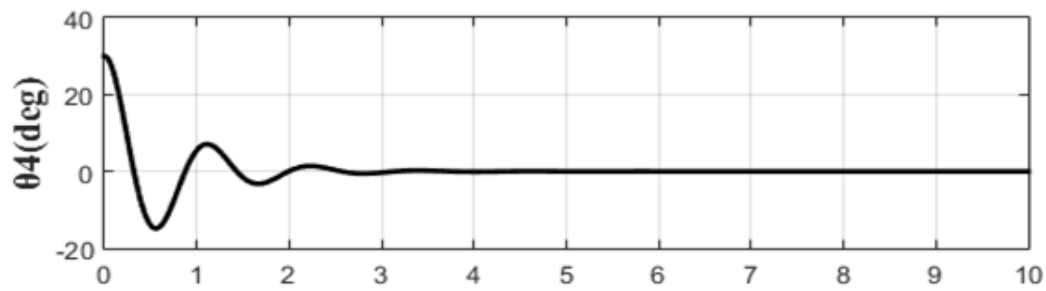


Figure 6. Displacement response of rod 4.

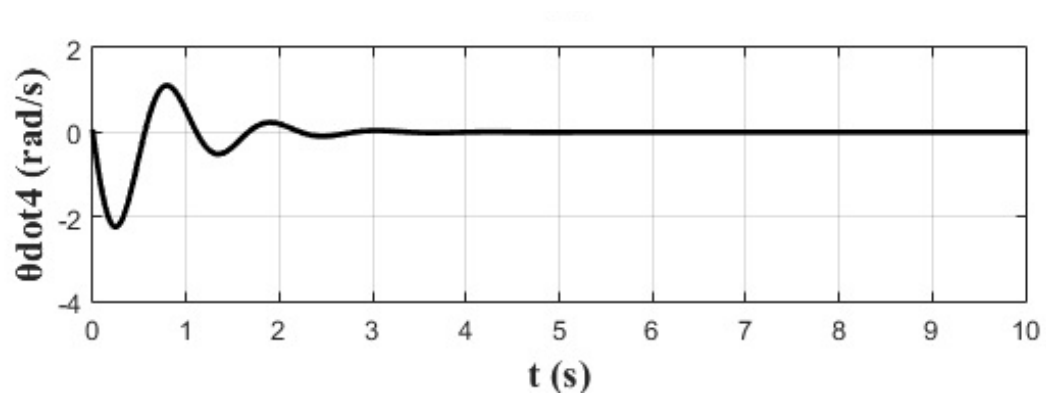


Figure 6b. Angular-velocity response of rod 4.

The responses shown above decay with time for the selected parameter set and initial condition. In the original project, this behavior is interpreted as asymptotic stability for all four degrees of freedom under the considered condition.

3. Small-Deviation Stability Analysis

The nonlinear equations are linearized about equilibrium configurations. Because the system is autonomous, local stability can be examined from the eigenvalues of the linearized state matrix. For asymptotic stability, the real parts of all eigenvalues must be negative.

3.1 Linearization about $\theta_1^* = \theta_2^* = \theta_3^* = \theta_4^* = 0$

$$\dot{x} = Ax + Bu.$$

Using the state vector $x = [\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2, \theta_3, \dot{\theta}_3, \theta_4, \dot{\theta}_4]^T$, the linearized matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -3(gL/2+k/m) & -3c/(mL^2) & 3k/m & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3k/m & 0 & -3(gL/2+2k/m) & -3c/(mL^2) & 3k/m & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3k/m & 0 & 0 & 0 & -3(gL/2+2k/m) & 0 \\ -3c/(mL^2) & 0 & 3k/m & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3(gL/2+k/m) & -3c/(mL^2) & 0 & 0 & 0 & 3k/m \end{bmatrix}$$

$$B = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ M]^T.$$

The source derives eight eigenvalues $s_1, \dots, s_8$. Their expressions are retained below in compact form:

$$s_1 = -[3c + \sqrt{3\sqrt{(3c^2 - 8kL^4m - 2gL^5m^2 - 4\sqrt{2} kL^4m)}}]/(2L^2m)$$

$$s_2 = -[3c + \sqrt{3\sqrt{(3c^2 - 8kL^4m - 2gL^5m^2 + 4\sqrt{2} kL^4m)}}]/(2L^2m)$$

$$s_3 = -[3c - \sqrt{3\sqrt{(3c^2 - 8kL^4m - 2gL^5m^2 - 4\sqrt{2} kL^4m)}}]/(2L^2m)$$

$$s_4 = -[3c - \sqrt{3\sqrt{(3c^2 - 8kL^4m - 2gL^5m^2 + 4\sqrt{2} kL^4m)}}]/(2L^2m)$$

$$s_5 = -[3c - 8\sqrt{(9c^2/64 - 3gL^5m^2/32)}]/(2L^2m)$$

$$s_6 = -[3c + 8\sqrt{(9c^2/64 - 3gL^5m^2/32)}]/(2L^2m)$$

$$s_7 = -[3c - 8\sqrt{(9c^2/64 - 3gL^5m^2/32 - 3kL^4m/8)}]/(2L^2m)$$

$$s_8 = -[3c + 8\sqrt{(9c^2/64 - 3gL^5m^2/32 - 3kL^4m/8)}]/(2L^2m)$$

For the numerical example used in the project, the state matrix becomes

$$A_1 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -4743/400 & -12/5 & 9/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 9/2 & 0 & -6543/400 & -12/5 & 9/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 9/2 & 0 & -6543/400 & -12/5 & 9/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 9/2 & 0 & -4743/400 & -12/5 & 0 & 0 \end{bmatrix}$$

The numerical eigenvalues reported in the source are

-1.2000	±	4.6132i
-1.2000	±	3.8623i
-1.2000	±	2.4326i
-1.2000 ± 2.9246i		

All reported eigenvalues have negative real parts; therefore, the equilibrium at $\theta = 0$ is locally asymptotically stable for the selected parameters.

3.2 Linearization about $\theta_1^* = \theta_2^* = \theta_3^* = \theta_4^* = \pi$

$\dot{x} = Fx + Bu$.

F				=
[	0	1	0	0
0			0	0
3(gL/2-k/m)	-3c/(mL ²)	3k/m	0	0
0		0		0
0	0	0	1	0
0		0		0
3k/m	0	3(gL/2-2k/m)	-3c/(mL ²)	3k/m
0		0		0
0	0	0	0	0
1		0		0
0	0	3k/m	0	3(gL/2-2k/m)
-3c/(mL ²)		3k/m		0
0	0	0	0	0
0		0		1
0	0	0	0	3k/m
0	3(gL/2-k/m)	-3c/(mL ²)	]	

The corresponding characteristic expressions reported in the source are

$$\begin{aligned}
 & -[3c + \sqrt{3}\sqrt{(3c^2 - 8kL^4m + 2gL^5m^2 - 4\sqrt{2}kL^4m)}]/(2L^2m) \\
 & -[3c + \sqrt{3}\sqrt{(3c^2 - 8kL^4m + 2gL^5m^2 + 4\sqrt{2}kL^4m)}]/(2L^2m) \\
 & -[3c - \sqrt{3}\sqrt{(3c^2 - 8kL^4m + 2gL^5m^2 - 4\sqrt{2}kL^4m)}]/(2L^2m) \\
 & -[3c - \sqrt{3}\sqrt{(3c^2 - 8kL^4m + 2gL^5m^2 + 4\sqrt{2}kL^4m)}]/(2L^2m) \\
 & -[3c - 8\sqrt{(9c^2/64 + 3gL^5m^2/32)}]/(2L^2m) \\
 & -[3c + 8\sqrt{(9c^2/64 + 3gL^5m^2/32)}]/(2L^2m) \\
 & -[3c - 8\sqrt{(9c^2/64 + 3gL^5m^2/32 - 3kL^4m/8)}]/(2L^2m) \\
 & -[3c + 8\sqrt{(9c^2/64 + 3gL^5m^2/32 - 3kL^4m/8)}]/(2L^2m)
 \end{aligned}$$

For the numerical example, the source reports

s ₁	=	+1.7661i
s ₂	=	+1.2822i
s ₃	=	-1.2000 + 2.5625i
s ₄	=	-1.2000 - 2.5625i
s ₅	=	-4.1661
s ₆	=	-3.6822
s ₇	=	-1.2000 + 0.4500i
s ₈	=	-1.2000 - 0.4500i

The reported eigenvalue set includes poles on the imaginary axis and the source therefore identifies the π equilibrium as unstable to small deviations. The large-deviation analysis is consequently not pursued about this equilibrium.

4. Large-Deviation Stability Analysis

The previous chapter examined local behavior using linearization. In this chapter, the nonlinear system is studied for larger deviations using Lyapunov's direct method. A natural candidate is the total mechanical energy, consisting of kinetic plus potential energy.

4.1 Lyapunov Function

$$V(q, \dot{q}) = T + U.$$

$$V = (kL^2/2)[(q_1 - q_2)^2 + (q_2 - q_3)^2 + (q_3 - q_4)^2] + (mL^2/6)(\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + \dot{q}_4^2) + (mgL/2)\sum_{i=1}^4(1 - \cos q_i).$$

The state variables used in the source are

$$y_1 = \theta_1, y_2 = \dot{\theta}_1, y_3 = \theta_2, y_4 = \dot{\theta}_2, y_5 = \theta_3, y_6 = \dot{\theta}_3, y_7 = \theta_4, y_8 = \dot{\theta}_4.$$

For the numerical parameters of the example, the Lyapunov function is written as

$$V = 3(y_1 - y_3)^2/32 + 3(y_3 - y_5)^2/32 + 3(y_5 - y_7)^2/32 - (981/800)[\cos y_1 + \cos y_3 + \cos y_5 + \cos y_7] + (y_2^2 + y_4^2 + y_6^2 + y_8^2)/48 + 981/200.$$

Along the system trajectories, the derivative of V reduces to the damping contribution

$$\dot{V} = -c(y_2^2 + y_4^2 + y_6^2 + y_8^2) \leq 0.$$

Thus V is non-increasing along trajectories. The source then uses a Lyapunov matrix construction to examine positive definiteness and the region of attraction.

4.2 Lyapunov Matrix Construction

The quadratic form is represented using

$$V = -x^T Q x.$$

Q		=
[0	0	0
0	0.1	0
0	0	0
0	0	0.1
0	0	0
0	0	0
0	0	0.1
0	0	0
0 0 0 0 0 0 0 0.1]		

The Lyapunov equation used in the numerical calculation is

$$A^T P + P A + Q = 0.$$

The source reports the following numerical P matrix:

P		≈
[0.0020	0	0.0006
0	0.0208	0
0.0006	0	0.0016
0	0	0.0208
0.0002	0	0.0005
0	0	0
0.0001	0	0.0002
0	0	0.0006
0	0	0.0208
0 0 0 0 0 0 0 0.0208]		

The source notes that values rounded to zero in the displayed matrix are retained numerically in MATLAB when plotting the attraction-domain surfaces. The reported eigenvalues of P are positive, so P is positive definite for the numerical example.

4.3 Lyapunov-Exponent Convergence

The original project uses a MATLAB Lyapunov-exponent toolbox with the system Jacobian and the quadratic-form information to compute the Lyapunov exponents. The plotted exponents converge toward zero, as shown in Figure 7.

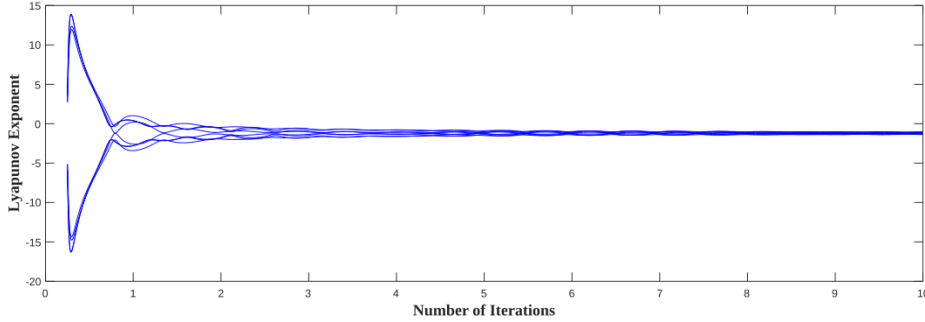


Figure 7. Convergence of the system's Lyapunov exponents.

4.4 Estimation of the Region of Attraction

To estimate the region of attraction, the minimum value of V on a level surface is considered. At a tangency point, the gradient of V is collinear with the normal to the corresponding level surface. Introducing a scalar multiplier β leads to the stationarity conditions used in the source.

$$\sin(y_1) = k(y_3 - y_1), \quad y_2 = -2c\beta y_2,$$

$$\sin(y_3) = k(y_1 + y_5 - 2y_3), \quad y_4 = -2c\beta y_4,$$

$$\sin(y_5) = k(y_3 + y_7 - 2y_5), \quad y_6 = -2c\beta y_6,$$

$$\sin(y_7) = k(y_5 - y_7), \quad y_8 = -2c\beta y_8,$$

$$y_2^2 + y_4^2 + y_6^2 + y_8^2 = 0.$$

The last condition implies $y_2 = y_4 = y_6 = y_8 = 0$. The remaining stationarity equations give the relevant equilibrium configurations. The source reports the corresponding V values as 0 and $m g L/4$ and uses the level

$$\Omega_k = \{x : V(x) < m g L/4\}.$$

The four plotted cross-sections of the attraction boundary are reproduced below.

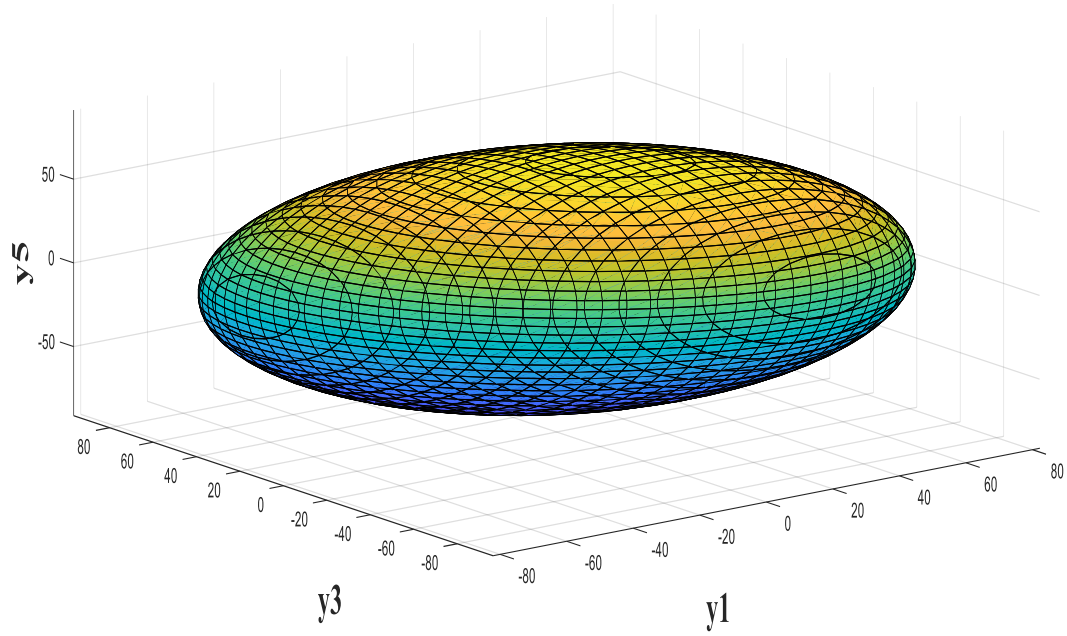


Figure 8. Lyapunov-function level surface for y_1 , y_3 , and y_5 .

Region 1: $0.0019831y_1^2 + 0.0011918y_1y_3 + 0.000366y_1y_5 + 0.00157y_3^2 + 0.000965y_3y_5 + 0.00157y_5^2 - 9.81 = 0$.

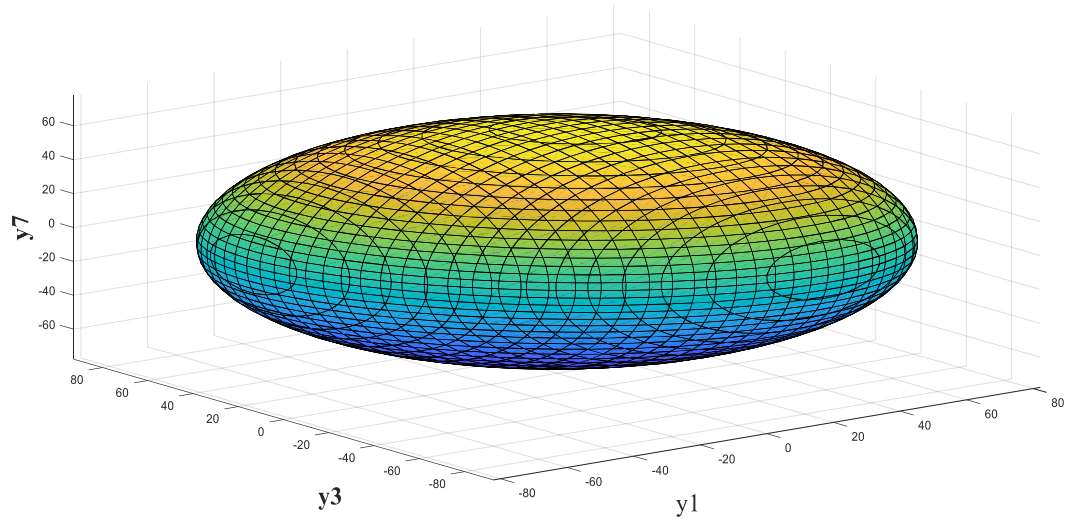


Figure 9. Lyapunov-function level surface for y_1 , y_3 , and y_7 .

Region 2: $0.0019831y_1^2 + 0.00119y_1y_3 + 0.000138y_1y_7 + 0.00157y_3^2 + 0.000366y_3y_7 + 0.00198y_7^2 - 9.81 = 0$.

Figure 10. Lyapunov-function level surface for y_1 , y_5 , and y_7 .

Region 3: $0.0019831y_1^2 + 0.000366y_1y_5 + 0.000138y_1y_7 + 0.00157y_5^2 + 0.00119y_5y_7 + 0.001983y_7^2 - 9.81 = 0$.

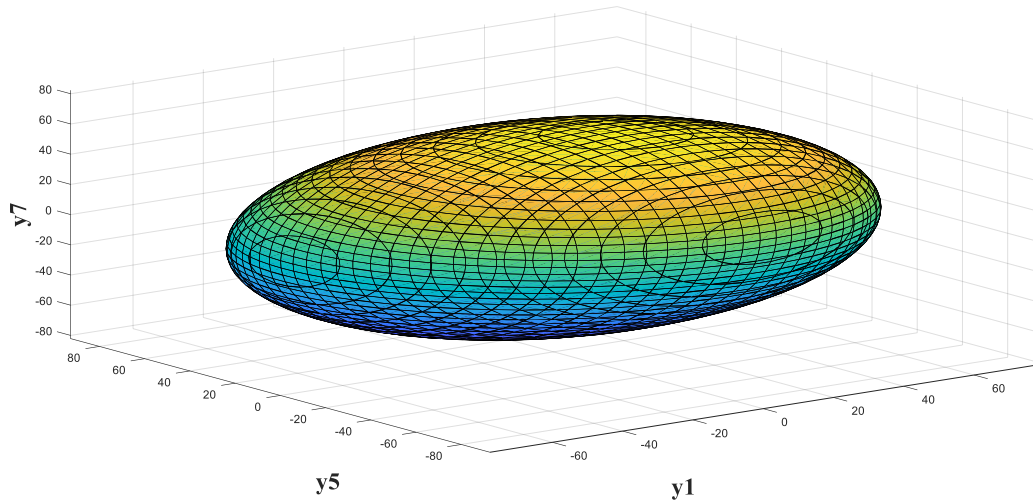
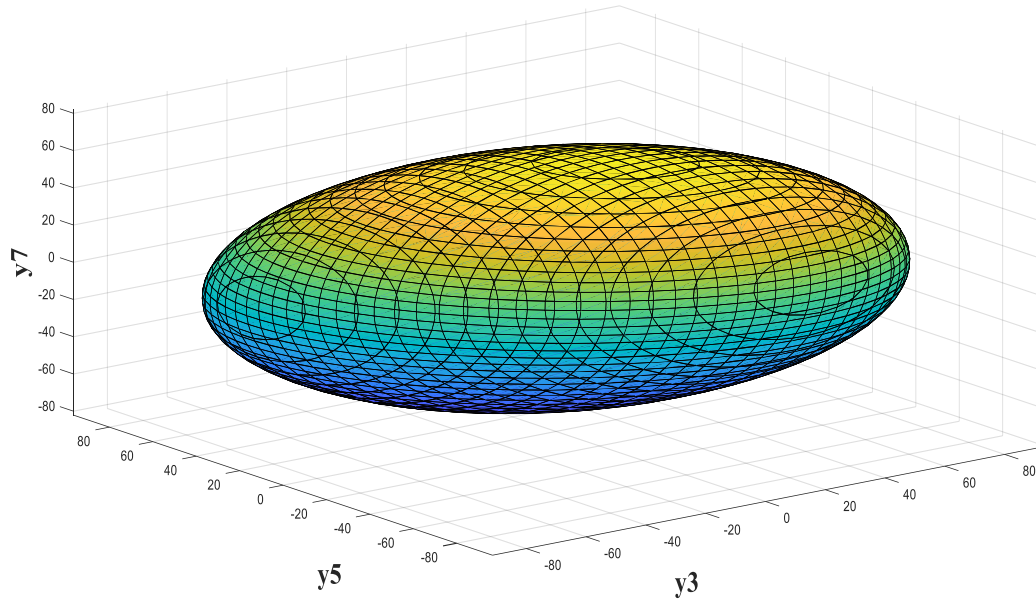


Figure 11. Lyapunov-function level surface for y_3 , y_5 , and y_7 .

Region 4: $0.00157y_3^2 + 0.000964y_3y_5 + 0.000366y_3y_7 + 0.00157y_5^2 + 0.00119y_5y_7 + 0.00198y_7^2 - 9.81 = 0$.

The source notes that the resulting three-dimensional boundaries have ellipsoidal-type cross-sections. Within the considered level set, V is used as an energy-like measure for the domain of attraction of the stable equilibrium.



5. Conclusions

A complete nonlinear model of the four-degree-of-freedom spring-coupled rod system was formulated using four classical mechanics approaches. Newton's equations, the Lagrange formulation, Hamilton's direct principle, and the canonical Hamilton formulation lead to consistent equations of motion.

For the numerical parameters and initial condition used in the original project, the simulated responses decay with time. Linearization about the $\theta = 0$ equilibrium gives eigenvalues with negative real parts, indicating local asymptotic stability for the selected parameter set.

In contrast, the π equilibrium exhibits eigenvalues on the imaginary axis in the reported numerical example, and the source identifies this equilibrium as unstable to small deviations.

For larger deviations about the stable equilibrium, the total mechanical energy provides a Lyapunov candidate. Its derivative is non-positive because of viscous damping. The Lyapunov-matrix calculation and the plotted level surfaces are then used to characterize a region of attraction for the nonlinear system.

References and Source Basis

This English project report is a faithful rewrite and typeset reconstruction of the supplied Persian project document. The terminology, parameter values, equations, numerical eigenvalues, figures, and reported conclusions are based on that source document; no external technical results were added.